

Creation Date 23-Dec-2010

Revision Date 20-Oct-2023

Revision Number 7

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

|                             |                                |
|-----------------------------|--------------------------------|
| <b>Product Description:</b> | <b>Orthophosphoric acid 1M</b> |
| <b>Cat No. :</b>            | <b>J/5655/PB08</b>             |
| <b>Synonyms</b>             | Phosphoric acid                |
| <b>Molecular Formula</b>    | H3PO4                          |

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

|                             |                          |
|-----------------------------|--------------------------|
| <b>Recommended Use</b>      | Laboratory chemicals.    |
| <b>Uses advised against</b> | No Information available |

### 1.3. Details of the supplier of the safety data sheet

#### Company

**EU entity/business name**  
Thermo Fisher Scientific  
Janssen Pharmaceuticaaan 3a  
2440 Geel, Belgium

**UK entity/business name**  
Fisher Scientific UK  
Bishop Meadow Road, Loughborough,  
Leicestershire LE11 5RG, United Kingdom

**Swiss distributor** - Fisher Scientific AG  
Neuhofstrasse 11, CH 4153 Reinach  
Tel: +41 (0) 56 618 41 11  
e-mail - infoch@thermofisher.com

**E-mail address** begel.sdsdesk@thermofisher.com

### 1.4. Emergency telephone number

Tel: 01509 231166  
Chemtrec US: (800) 424-9300  
Chemtrec EU: 001-703-527-3887

For customers in Switzerland:  
Tox Info Suisse Emergency Number: **145 (24hr)**  
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)  
Chemtrec (24h) Toll-Free: 0800 564 402  
Chemtrec Local: +41-43 508 20 11 (Zurich)

## SECTION 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

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## CLP Classification - Regulation (EC) No 1272/2008

### Physical hazards

Based on available data, the classification criteria are not met

### Health hazards

Based on available data, the classification criteria are not met

### Environmental hazards

Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

## 2.2. Label elements

None required

## 2.3. Other hazards

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

### 3.2. Mixtures

| Component       | CAS No    | EC No             | Weight % | CLP Classification - Regulation (EC) No 1272/2008                                       |
|-----------------|-----------|-------------------|----------|-----------------------------------------------------------------------------------------|
| Phosphoric acid | 7664-38-2 | EEC No. 231-633-2 | 8 - 10   | Met. Corr. 1 (H290)<br>Acute Tox. 4 (H302)<br>Skin Corr. 1B (H314)<br>Eye Dam. 1 (H318) |
| Water           | 7732-18-5 | 231-791-2         | 90 - 92  | -                                                                                       |

| Component       | Specific concentration limits (SCL's)                                                | M-Factor | Component notes |
|-----------------|--------------------------------------------------------------------------------------|----------|-----------------|
| Phosphoric acid | Skin Corr. 1B :: C>=25%<br>Eye Irrit. 2 :: 10%<=C<25%<br>Skin Irrit. 2 :: 10%<=C<25% | -        | -               |

| Components           | Reach Registration Number |
|----------------------|---------------------------|
| Orthophosphoric acid | 01-2119485924-24          |

Full text of Hazard Statements: see section 16

## SECTION 4: FIRST AID MEASURES

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## 4.1. Description of first aid measures

|                                           |                                                                                                                 |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention. |
| <b>Skin Contact</b>                       | Wash off immediately with plenty of water for at least 15 minutes. Get medical attention.                       |
| <b>Ingestion</b>                          | Do NOT induce vomiting. Get medical attention.                                                                  |
| <b>Inhalation</b>                         | Remove to fresh air. If breathing is difficult, give oxygen. Get medical attention.                             |
| <b>Self-Protection of the First Aider</b> | No special precautions required.                                                                                |

## 4.2. Most important symptoms and effects, both acute and delayed

No information available.

## 4.3. Indication of any immediate medical attention and special treatment needed

|                           |                        |
|---------------------------|------------------------|
| <b>Notes to Physician</b> | Treat symptomatically. |
|---------------------------|------------------------|

## SECTION 5: FIREFIGHTING MEASURES

### 5.1. Extinguishing media

#### **Suitable Extinguishing Media**

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

#### **Extinguishing media which must not be used for safety reasons**

No information available.

### 5.2. Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors.

#### **Hazardous Combustion Products**

Thermal decomposition can lead to release of irritating gases and vapors, Oxides of phosphorus.

### 5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### 6.1. Personal precautions, protective equipment and emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required.

### 6.2. Environmental precautions

Should not be released into the environment. Do not flush into surface water or sanitary sewer system. See Section 12 for additional Ecological Information.

### 6.3. Methods and material for containment and cleaning up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

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## 6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### 7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Wash hands before breaks and immediately after handling the product.

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

### 7.2. Conditions for safe storage, including any incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place.

**Technical Rules for Hazardous Substances (TRGS) 510**  
**Storage Class (LGK) (Germany)**

Storage Class/LGK 12

**Switzerland - Storage of hazardous substances**

Storage class - SC 10/12  
<https://www.kvu.ch/de/themen/stoffe-und-produkte>  
<https://www.kvu.ch/fr/themes/substances-et-produits>  
<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

### 7.3. Specific end use(s)

Use in laboratories

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### 8.1. Control parameters

#### Exposure limits

List source(s): **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE** - 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

| Component       | European Union                                                     | The United Kingdom                                    | France                                                                                                                                                                                                                | Belgium                                                                 | Spain                                                                                           |
|-----------------|--------------------------------------------------------------------|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Phosphoric acid | TWA: 1 mg/m <sup>3</sup> (8h)<br>STEL: 2 mg/m <sup>3</sup> (15min) | STEL: 2 mg/m <sup>3</sup><br>TWA: 1 mg/m <sup>3</sup> | TWA / VME: 0.2 ppm (8 heures). indicative limit<br>TWA / VME: 1 mg/m <sup>3</sup> (8 heures). indicative limit<br>STEL / VLCT: 0.5 ppm.<br>indicative limit<br>STEL / VLCT: 2 mg/m <sup>3</sup> .<br>indicative limit | TWA: 1 mg/m <sup>3</sup> 8 uren<br>STEL: 2 mg/m <sup>3</sup> 15 minuten | STEL / VLA-EC: 2 mg/m <sup>3</sup> (15 minutos).<br>TWA / VLA-ED: 1 mg/m <sup>3</sup> (8 horas) |

| Component       | Italy                                                                                                       | Germany                                                                                                    | Portugal                                                                 | The Netherlands                                                         | Finland                                                                         |
|-----------------|-------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|-------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Phosphoric acid | TWA: 1 mg/m <sup>3</sup> 8 ore.<br>Time Weighted Average<br>STEL: 2 mg/m <sup>3</sup> 15 minuti. Short-term | TWA: 2 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 2<br>TWA: 2 mg/m <sup>3</sup> (8 Stunden). MAK | STEL: 2 mg/m <sup>3</sup> 15 minutos<br>TWA: 1 mg/m <sup>3</sup> 8 horas | STEL: 2 mg/m <sup>3</sup> 15 minuten<br>TWA: 1 mg/m <sup>3</sup> 8 uren | TWA: 1 mg/m <sup>3</sup> 8 tunteina<br>STEL: 2 mg/m <sup>3</sup> 15 minuutteina |

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|                 |                                                                                                   | Höhepunkt: 4 mg/m <sup>3</sup>                                                              |                                                                                                                        |                                                                                              |                                                                                                   |
|-----------------|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Component       | Austria                                                                                           | Denmark                                                                                     | Switzerland                                                                                                            | Poland                                                                                       | Norway                                                                                            |
| Phosphoric acid | MAK-KZGW: 2 mg/m <sup>3</sup><br>15 Minuten<br>MAK-TMW: 1 mg/m <sup>3</sup> 8<br>Stunden          | TWA: 1 mg/m <sup>3</sup> 8 timer<br>STEL: 2 mg/m <sup>3</sup> 15<br>minutter                | STEL: 4 mg/m <sup>3</sup> 15<br>Minuten<br>TWA: 2 mg/m <sup>3</sup> 8<br>Stunden                                       | STEL: 2 mg/m <sup>3</sup> 15<br>minutach<br>TWA: 1 mg/m <sup>3</sup> 8<br>godzinach          | TWA: 1 mg/m <sup>3</sup> 8 timer<br>STEL: 3 mg/m <sup>3</sup> 15<br>minutter. value<br>calculated |
| Component       | Bulgaria                                                                                          | Croatia                                                                                     | Ireland                                                                                                                | Cyprus                                                                                       | Czech Republic                                                                                    |
| Phosphoric acid | TWA: 1.0 mg/m <sup>3</sup><br>STEL : 2.0 mg/m <sup>3</sup>                                        | TWA-GVI: 1 mg/m <sup>3</sup> 8<br>satima.<br>STEL-KGVI: 2 mg/m <sup>3</sup> 15<br>minutama. | TWA: 1 mg/m <sup>3</sup> 8 hr.<br>STEL: 2 mg/m <sup>3</sup> 15 min                                                     | STEL: 2.0 mg/m <sup>3</sup><br>TWA: 1 mg/m <sup>3</sup>                                      | TWA: 1 mg/m <sup>3</sup> 8<br>hodinách.<br>Ceiling: 2 mg/m <sup>3</sup>                           |
| Component       | Estonia                                                                                           | Gibraltar                                                                                   | Greece                                                                                                                 | Hungary                                                                                      | Iceland                                                                                           |
| Phosphoric acid | TWA: 1 mg/m <sup>3</sup> 8<br>tundides. vapor<br>STEL: 2 mg/m <sup>3</sup> 15<br>minutites. vapor | TWA: 1 mg/m <sup>3</sup> 8 hr<br>STEL: 2 mg/m <sup>3</sup> 15 min                           | STEL: 3 mg/m <sup>3</sup><br>TWA: 1 mg/m <sup>3</sup>                                                                  | STEL: 2 mg/m <sup>3</sup> 15<br>percekben. CK<br>TWA: 1 mg/m <sup>3</sup> 8<br>óraban. AK    | STEL: 2 mg/m <sup>3</sup><br>TWA: 1 mg/m <sup>3</sup> 8<br>klukkustundum.                         |
| Component       | Latvia                                                                                            | Lithuania                                                                                   | Luxembourg                                                                                                             | Malta                                                                                        | Romania                                                                                           |
| Phosphoric acid | STEL: 2 mg/m <sup>3</sup><br>TWA: 1 mg/m <sup>3</sup>                                             | TWA: 1 mg/m <sup>3</sup> IPRD<br>STEL: 2 mg/m <sup>3</sup>                                  | TWA: 1 mg/m <sup>3</sup> 8<br>Stunden<br>STEL: 2 mg/m <sup>3</sup> 15<br>Minuten                                       | TWA: 1 mg/m <sup>3</sup><br>STEL: 2 mg/m <sup>3</sup> 15<br>minuti                           | TWA: 1 mg/m <sup>3</sup> 8 ore<br>STEL: 2 mg/m <sup>3</sup> 15<br>minute                          |
| Component       | Russia                                                                                            | Slovak Republic                                                                             | Slovenia                                                                                                               | Sweden                                                                                       | Turkey                                                                                            |
| Phosphoric acid |                                                                                                   | Ceiling: 2 mg/m <sup>3</sup><br>TWA: 1 mg/m <sup>3</sup>                                    | TWA: 1 mg/m <sup>3</sup> 8 urah<br>inhalable fraction<br>STEL: 2 mg/m <sup>3</sup> 15<br>minutah inhalable<br>fraction | Binding STEL: 2 mg/m <sup>3</sup><br>15 minuter<br>TLV: 1 mg/m <sup>3</sup> 8 timmar.<br>NGV | TWA: 1 mg/m <sup>3</sup> 8 saat<br>STEL: 2 mg/m <sup>3</sup> 15<br>dakika                         |

## Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

## Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

## Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

| Component                               | Acute effects local<br>(Dermal) | Acute effects<br>systemic (Dermal) | Chronic effects local<br>(Dermal) | Chronic effects<br>systemic (Dermal) |
|-----------------------------------------|---------------------------------|------------------------------------|-----------------------------------|--------------------------------------|
| Phosphoric acid<br>7664-38-2 ( 8 - 10 ) |                                 | DNEL = 134.5mg/kg<br>bw/day        |                                   | DNEL = 3.8mg/kg<br>bw/day            |

| Component                               | Acute effects local<br>(Inhalation) | Acute effects<br>systemic (Inhalation) | Chronic effects local<br>(Inhalation) | Chronic effects<br>systemic (Inhalation) |
|-----------------------------------------|-------------------------------------|----------------------------------------|---------------------------------------|------------------------------------------|
| Phosphoric acid<br>7664-38-2 ( 8 - 10 ) | DNEL = 1mg/m <sup>3</sup>           | DNEL = 948.6mg/m <sup>3</sup>          | DNEL = 1mg/m <sup>3</sup>             | DNEL = 13.2mg/m <sup>3</sup>             |

## Predicted No Effect Concentration (PNEC)

See values below.

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| Component                               | Fresh water    | Fresh water sediment           | Water Intermittent | Microorganisms in sewage treatment | Soil (Agriculture)          |
|-----------------------------------------|----------------|--------------------------------|--------------------|------------------------------------|-----------------------------|
| Phosphoric acid<br>7664-38-2 ( 8 - 10 ) | PNEC = 100µg/L | PNEC = 392µg/kg<br>sediment dw | PNEC = 1000µg/L    | PNEC = 100mg/L                     | PNEC = 19.7µg/kg<br>soil dw |

| Component                               | Marine water  | Marine water sediment           | Marine water Intermittent | Food chain            | Air |
|-----------------------------------------|---------------|---------------------------------|---------------------------|-----------------------|-----|
| Phosphoric acid<br>7664-38-2 ( 8 - 10 ) | PNEC = 10µg/L | PNEC = 39.2µg/kg<br>sediment dw |                           | PNEC = 4mg/kg<br>food |     |

## 8.2. Exposure controls

### Engineering Measures

None under normal use conditions.

### Personal protective equipment

#### Eye Protection

Wear safety glasses with side shields (or goggles) (European standard - EN 166)

#### Hand Protection

Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Natural rubber | See manufacturers | -               | EN 374      | (minimum requirement) |
| Nitrile rubber | recommendations   |                 |             |                       |
| Neoprene       |                   |                 |             |                       |
| PVC            |                   |                 |             |                       |

#### Skin and body protection

Wear appropriate protective gloves and clothing to prevent skin exposure.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

#### Respiratory Protection

No protective equipment is needed under normal use conditions.

#### Large scale/emergency use

Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Particle filter

#### Small scale/Laboratory use

Maintain adequate ventilation

#### Environmental exposure controls

Prevent product from entering drains. Do not allow material to contaminate ground water system.

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1. Information on basic physical and chemical properties

|                     |                          |
|---------------------|--------------------------|
| Physical State      | Liquid                   |
| Appearance          | Clear                    |
| Odor                | No information available |
| Odor Threshold      | No data available        |
| Melting Point/Range | No data available        |
| Softening Point     | No data available        |
| Boiling Point/Range | No information available |

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|                                         |                          |                                          |
|-----------------------------------------|--------------------------|------------------------------------------|
| Flammability (liquid)                   | No data available        |                                          |
| Flammability (solid,gas)                | Not applicable           | Liquid                                   |
| Explosion Limits                        | No data available        |                                          |
| Flash Point                             | No information available | <b>Method -</b> No information available |
| Autoignition Temperature                | No data available        |                                          |
| Decomposition Temperature               | No data available        |                                          |
| pH                                      |                          | Acidic                                   |
| Viscosity                               | No data available        |                                          |
| Water Solubility                        | Miscible                 |                                          |
| Solubility in other solvents            | No information available |                                          |
| Partition Coefficient (n-octanol/water) |                          |                                          |
| Vapor Pressure                          | 0.03 mmHg @ 20 °C        |                                          |
| Density / Specific Gravity              | 1.054                    |                                          |
| Bulk Density                            | Not applicable           | Liquid                                   |
| Vapor Density                           | No data available        | (Air = 1.0)                              |
| Particle characteristics                | Not applicable (liquid)  |                                          |

## 9.2. Other information

|                   |       |
|-------------------|-------|
| Molecular Formula | H3PO4 |
| Molecular Weight  | 98.00 |

## SECTION 10: STABILITY AND REACTIVITY

### 10.1. Reactivity

None known, based on information available

### 10.2. Chemical stability

Stable under normal conditions.

### 10.3. Possibility of hazardous reactions

|                          |                                          |
|--------------------------|------------------------------------------|
| Hazardous Polymerization | Hazardous polymerization does not occur. |
| Hazardous Reactions      | Corrosive to metals.                     |

### 10.4. Conditions to avoid

Incompatible products. Excess heat.

### 10.5. Incompatible materials

Metals. Strong bases.

### 10.6. Hazardous decomposition products

Thermal decomposition can lead to release of irritating gases and vapors. Oxides of phosphorus.

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

|                     |                                                     |
|---------------------|-----------------------------------------------------|
| Product Information | See actual entry in RTECS for complete information. |
|---------------------|-----------------------------------------------------|

#### (a) acute toxicity;

Oral

Dermal

Inhalation

Based on available data, the classification criteria are not met

Based on available data, the classification criteria are not met

Based on available data, the classification criteria are not met

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## Toxicology data for the components

| Component       | LD50 Oral                 | LD50 Dermal                  | LC50 Inhalation                   |
|-----------------|---------------------------|------------------------------|-----------------------------------|
| Phosphoric acid | LD50 = 1530 mg/kg ( Rat ) | LD50 = 2740 mg/kg ( Rabbit ) | 850 mg/m <sup>3</sup> ( Rat ) 1 h |
| Water           | -                         | -                            | -                                 |

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

(d) respiratory or skin sensitization;

Respiratory No data available

Skin No data available

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available

There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs No information available.

(j) aspiration hazard; No data available

Symptoms / effects, both acute and delayed No information available.

## 11.2. Information on other hazards

**Endocrine Disrupting Properties** Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

### 12.1. Toxicity

#### **Ecotoxicity effects**

Contains no substances known to be hazardous to the environment or that are not degradable in waste water treatment plants.

| Component       | Freshwater Fish         | Water Flea             | Freshwater Algae |
|-----------------|-------------------------|------------------------|------------------|
| Phosphoric acid | 98 - 106 mg/L LC50 96 h | > 100 mg/L EC50 = 48 h |                  |



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## 12.2. Persistence and degradability

### **Persistence**

Miscible with water, Persistence is unlikely, based on information available.

## 12.3. Bioaccumulative potential

Bioaccumulation is unlikely

## 12.4. Mobility in soil

The product is water soluble, and may spread in water systems Will likely be mobile in the environment due to its water solubility. Highly mobile in soils

## 12.5. Results of PBT and vPvB assessment

No data available for assessment.

## 12.6. Endocrine disrupting properties

### **Endocrine Disruptor Information**

This product does not contain any known or suspected endocrine disruptors

## 12.7. Other adverse effects

### **Persistent Organic Pollutant Ozone Depletion Potential**

This product does not contain any known or suspected substance

This product does not contain any known or suspected substance

## **SECTION 13: DISPOSAL CONSIDERATIONS**

### 13.1. Waste treatment methods

#### **Waste from Residues/Unused Products**

Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

#### **Contaminated Packaging**

Empty remaining contents. Dispose of in accordance with local regulations. Do not re-use empty containers.

#### **European Waste Catalogue (EWC)**

According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

#### **Other Information**

Do not flush to sewer.

#### **Switzerland - Waste Ordinance**

Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600  
<https://www.fedlex.admin.ch/eli/cc/2015/891/en>

## **SECTION 14: TRANSPORT INFORMATION**

### IMDG/IMO

Not regulated

#### 14.1. UN number

#### 14.2. UN proper shipping name

#### 14.3. Transport hazard class(es)

#### 14.4. Packing group

### ADR

Not regulated

#### 14.1. UN number

#### 14.2. UN proper shipping name

#### 14.3. Transport hazard class(es)

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## 14.4. Packing group

IATA Not regulated

### 14.1. UN number

### 14.2. UN proper shipping name

### 14.3. Transport hazard class(es)

### 14.4. Packing group

14.5. Environmental hazards No hazards identified

14.6. Special precautions for user No special precautions required.

14.7. Maritime transport in bulk according to IMO instruments Not applicable, packaged goods

## SECTION 15: REGULATORY INFORMATION

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component       | CAS No    | EINECS    | ELINCS | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|-----------------|-----------|-----------|--------|-----|-------|------|----------|------|------|
| Phosphoric acid | 7664-38-2 | 231-633-2 | -      | -   | X     | X    | KE-27427 | X    | X    |
| Water           | 7732-18-5 | 231-791-2 | -      | -   | X     | X    | KE-35400 | X    | -    |

| Component       | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|-----------------|-----------|------|-----------------------------------------------|-----|------|------|-------|-------|
| Phosphoric acid | 7664-38-2 | X    | ACTIVE                                        | X   | -    | X    | X     | X     |
| Water           | 7732-18-5 | X    | ACTIVE                                        | X   | -    | X    | X     | X     |

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

#### Authorisation/Restrictions according to EU REACH

| Component       | CAS No    | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------------|-----------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Phosphoric acid | 7664-38-2 | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |
| Water           | 7732-18-5 | -                                                                   | -                                                                             | -                                                                                                     |

#### REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

#### Seveso III Directive (2012/18/EC)

| Component       | CAS No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|-----------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Phosphoric acid | 7664-38-2 | Not applicable                                                                            | Not applicable                                                                           |
| Water           | 7732-18-5 | Not applicable                                                                            | Not applicable                                                                           |

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Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

## National Regulations

UK - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

## WGK Classification

Water endangering class = 1 (self classification)

| Component       | Germany - Water Classification (AwSV) | Germany - TA-Luft Class |
|-----------------|---------------------------------------|-------------------------|
| Phosphoric acid | WGK1                                  |                         |

## Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

| Component                               | Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81) | Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC) | Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Phosphoric acid<br>7664-38-2 ( 8 - 10 ) | Prohibited and Restricted Substances                                                                           |                                                                                 |                                                                                             |

## 15.2. Chemical safety assessment

Chemical Safety Assessment/Reports (CSA/CSR) are not required for mixtures

## SECTION 16: OTHER INFORMATION

### Full text of H-Statements referred to under sections 2 and 3

H314 - Causes severe skin burns and eye damage

H318 - Causes serious eye damage

H290 - May be corrosive to metals

### Legend

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

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**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (volatile organic compound)

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

## Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

**Physical hazards** On basis of test data

**Health Hazards** Calculation method

**Environmental hazards** Calculation method

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

**Creation Date** 23-Dec-2010

**Revision Date** 20-Oct-2023

**Revision Summary** Not applicable.

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006. COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No 1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2, Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and Preparations).**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**